AWS VPC Peering Project

EWB Internship Project Report

Introduction

In this project, I configured two Amazon Virtual Private Clouds (VPCs), created public and private subnets, set up route tables and security groups, launched EC2 instances, and connected the VPCs through VPC peering. I used SSH to access the private instance through a public jump host and tested connectivity between private instances using their private IP addresses.

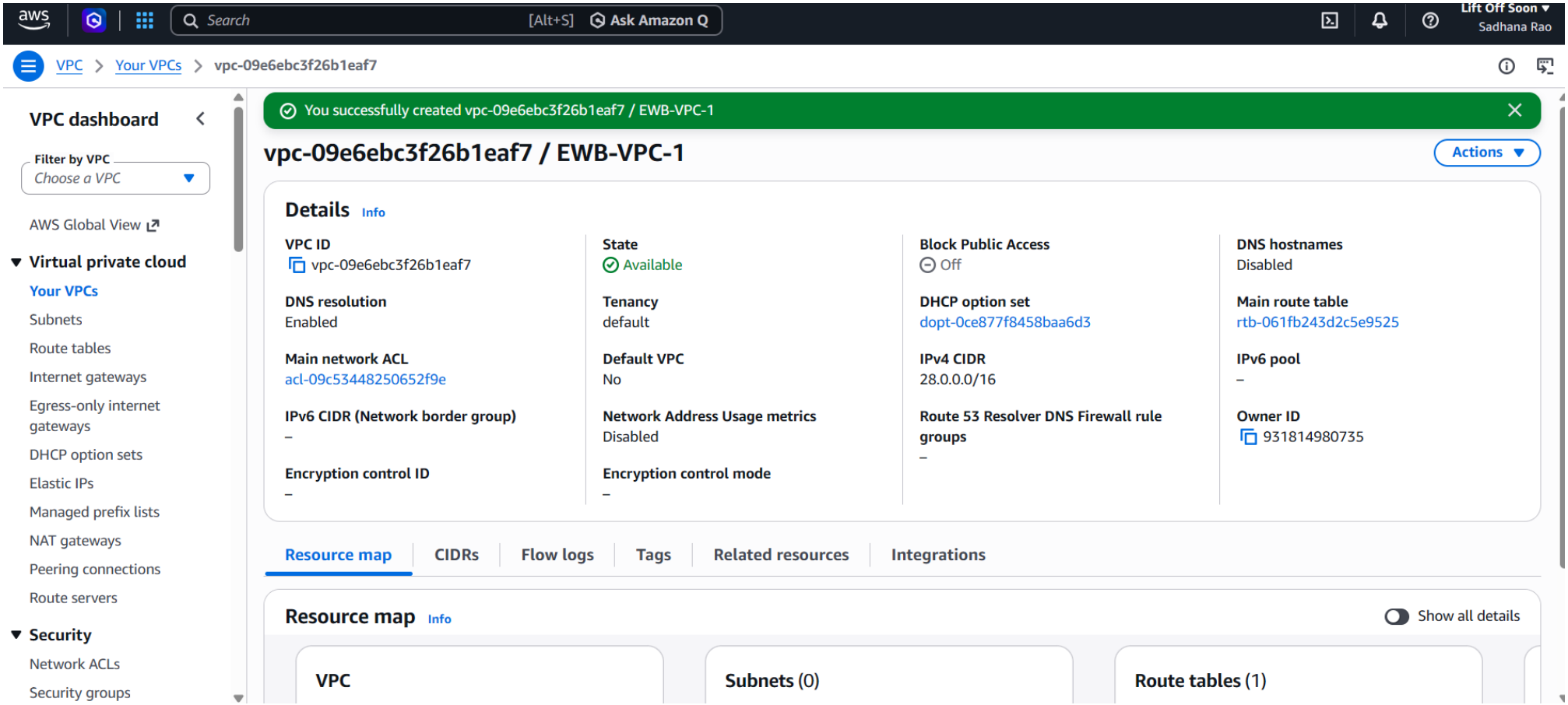
Network Configuration

VPC 1: 28.0.0.0/16 | **Public subnet:** 28.0.1.0/24 | **Private subnet:** 28.0.2.0/24
VPC 2: 28.200.0.0/16 | **Private subnet:** 28.200.1.0/24

The screenshots in this report are arranged in project workflow order. Each image has a descriptive title and a short explanation of the configuration step it documents.

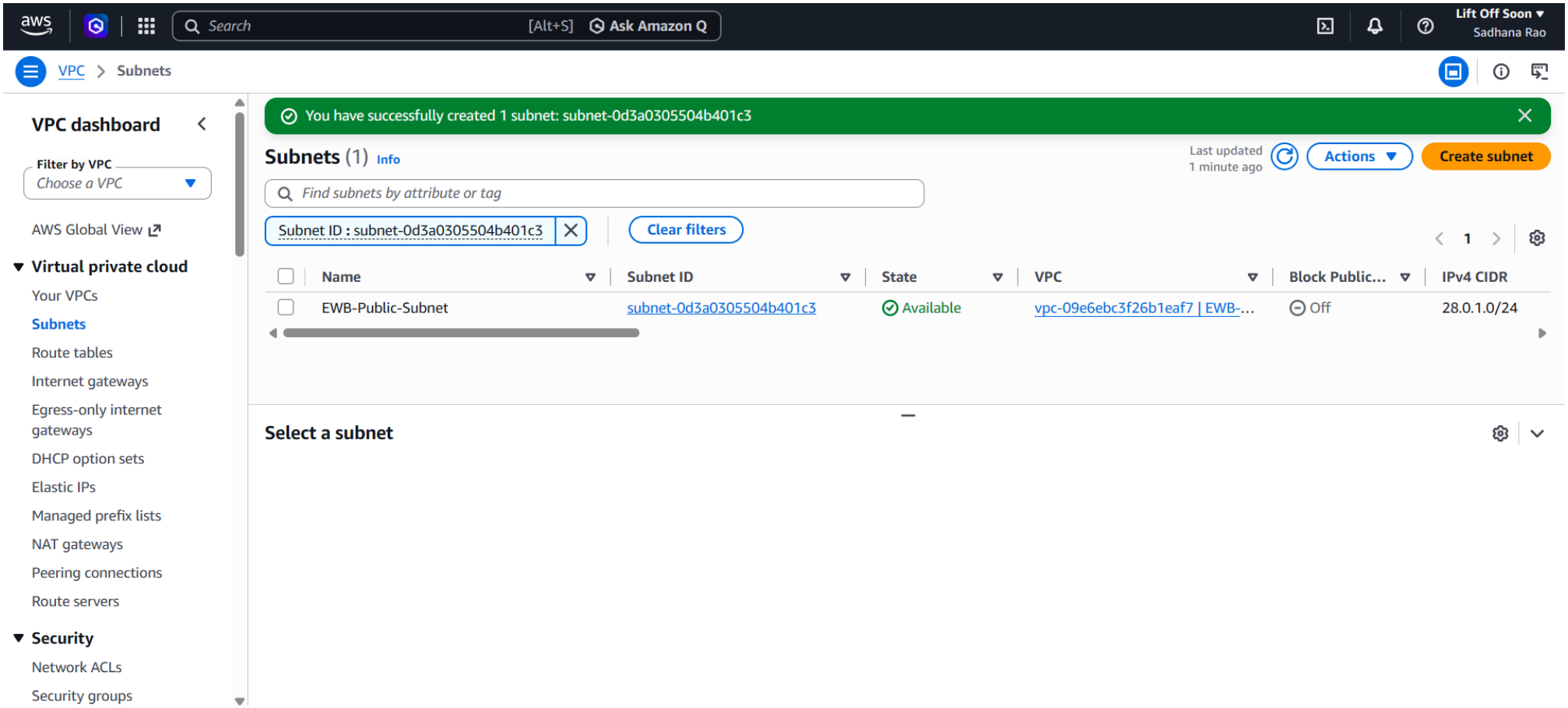
01. Create VPC 1 — EWB-VPC-1

Create the first VPC with CIDR block 28.0.0.0/16.



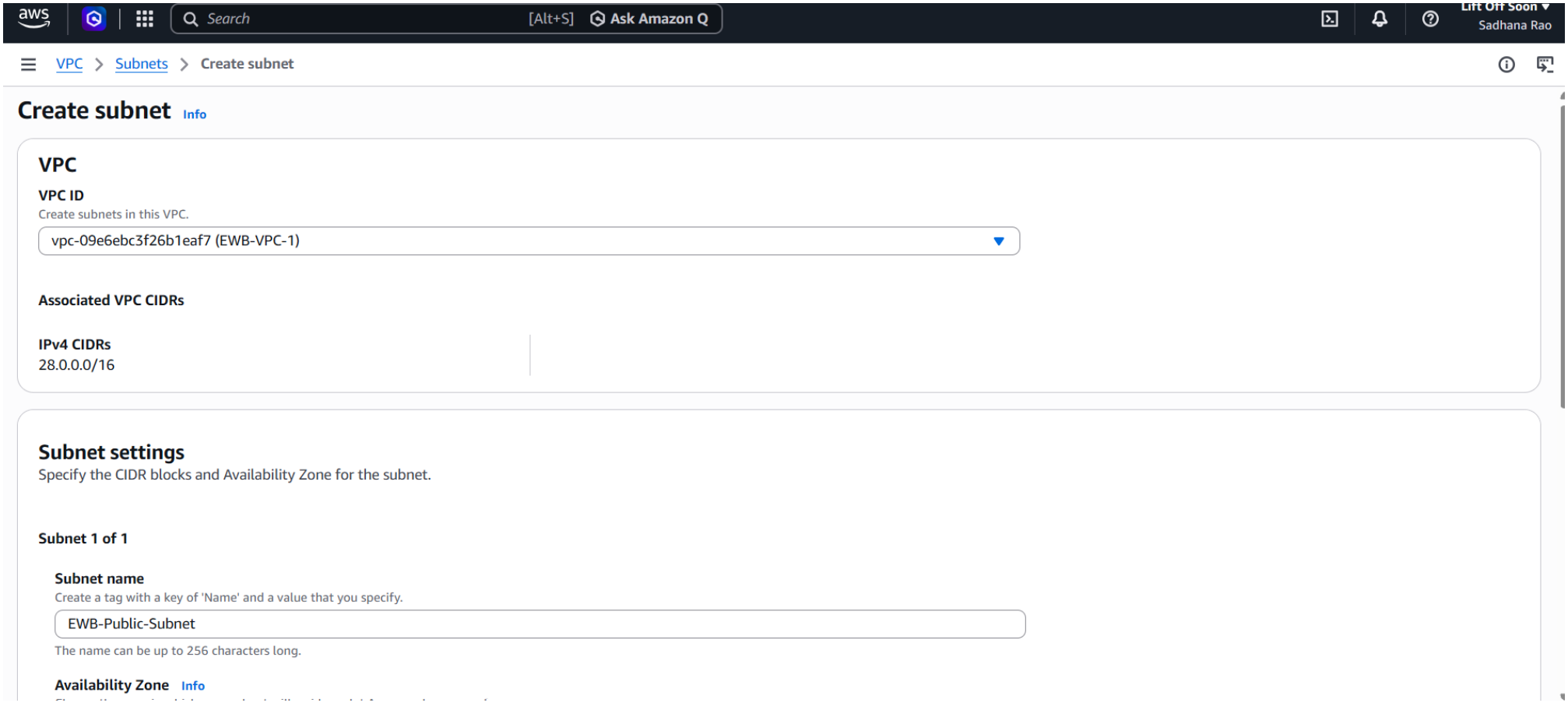
02. Create the Public Subnet in VPC 1

Configure EWB-Public-Subnet with CIDR block 28.0.1.0/24.



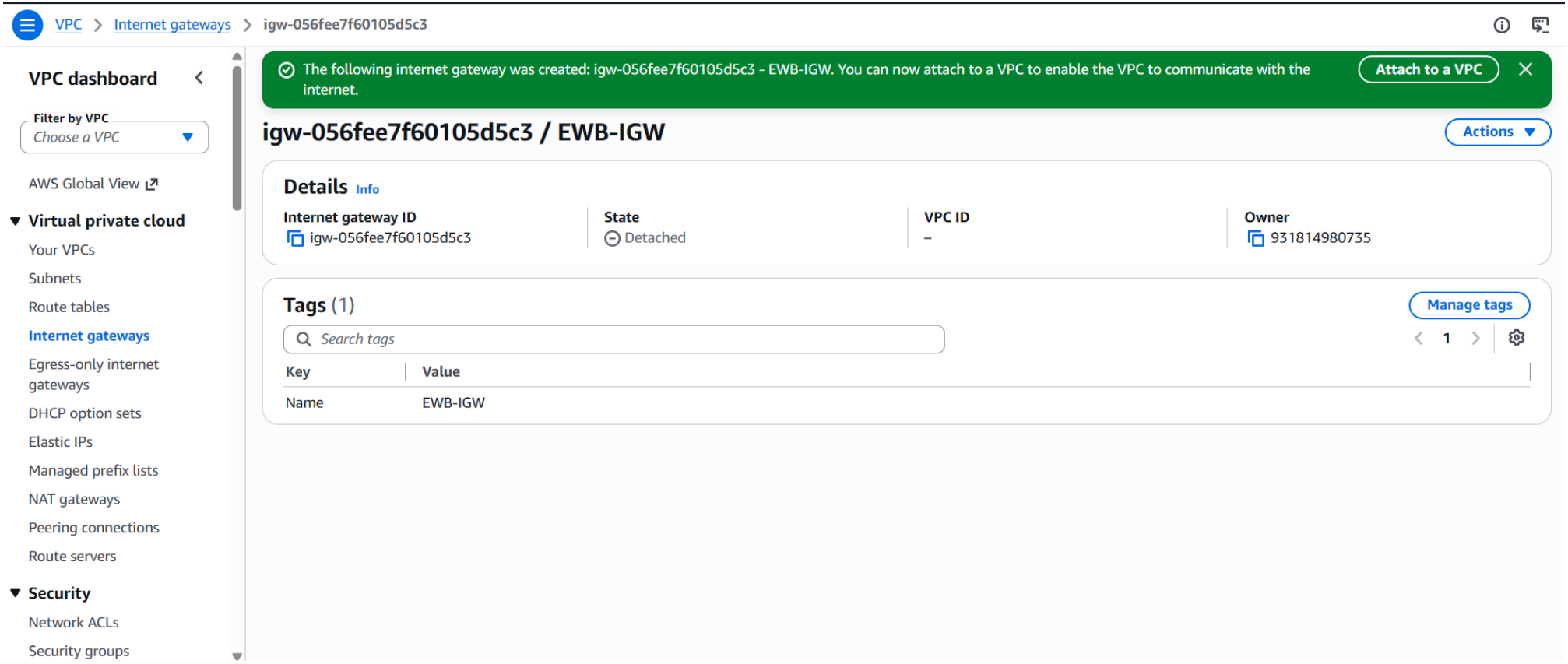
03. Create the Private Subnet in VPC 1

Configure EWB-Private-Subnet with CIDR block 28.0.2.0/24.



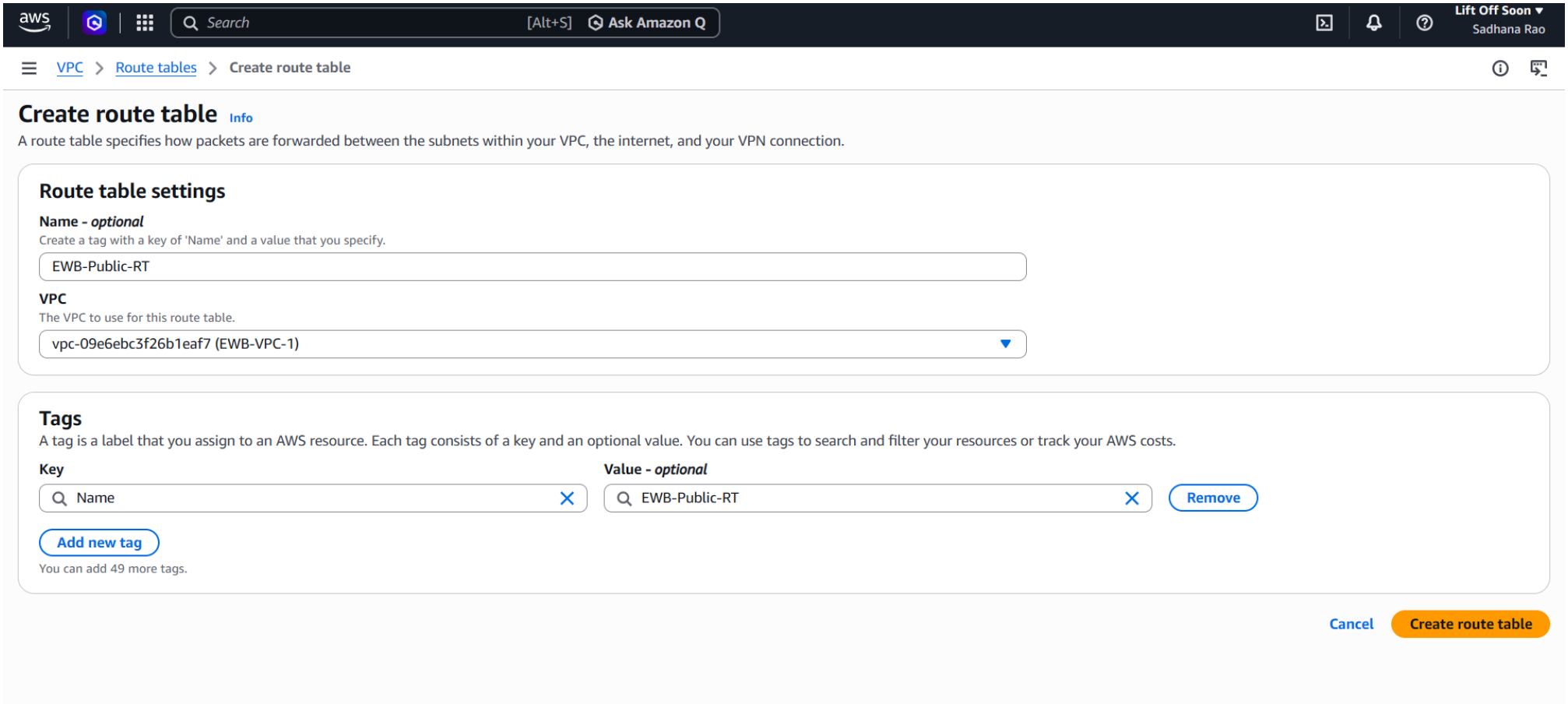
04. Create and Attach the Internet Gateway

Create EWB-IGW and attach it to EWB-VPC-1.



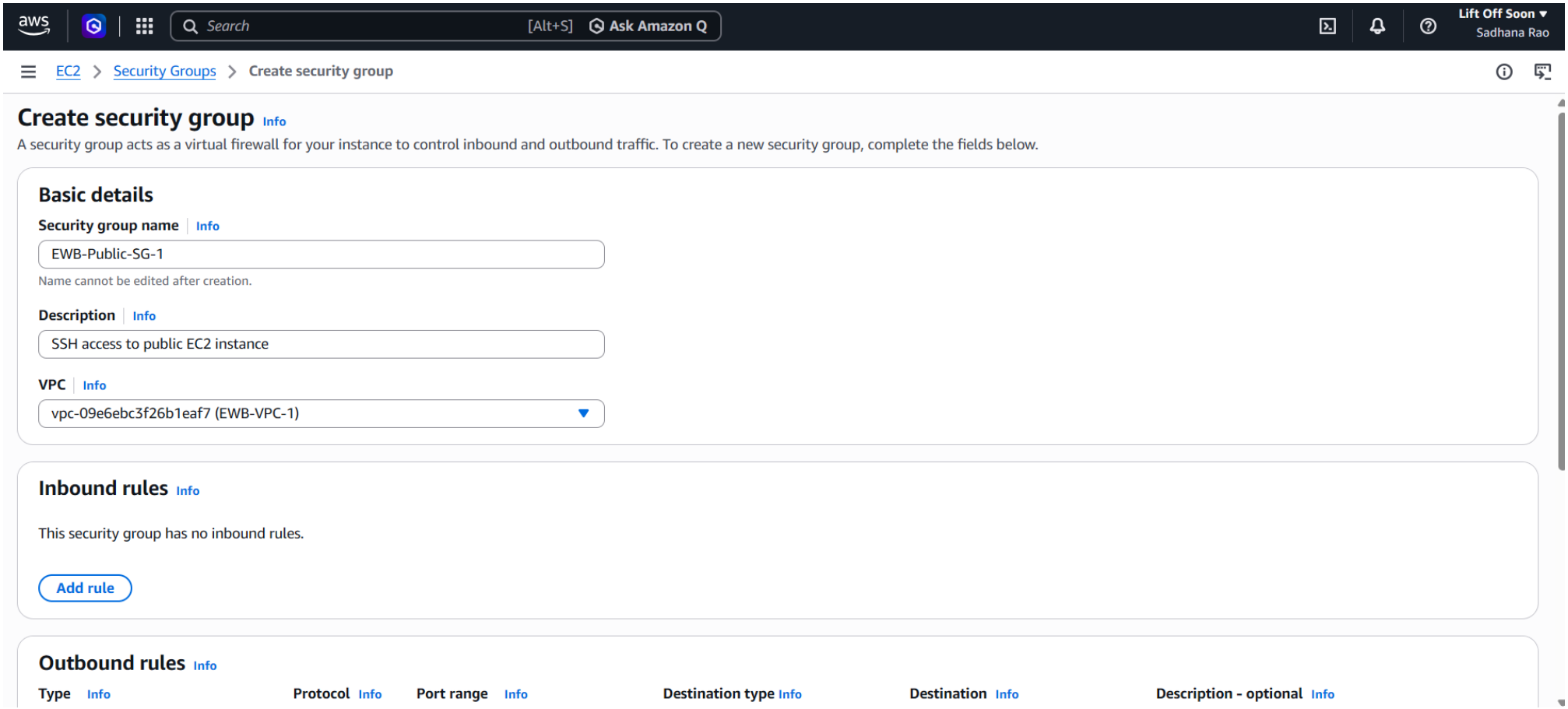
05. Configure the Route Table for VPC 1

Create the route table and configure the routes needed for subnet traffic.



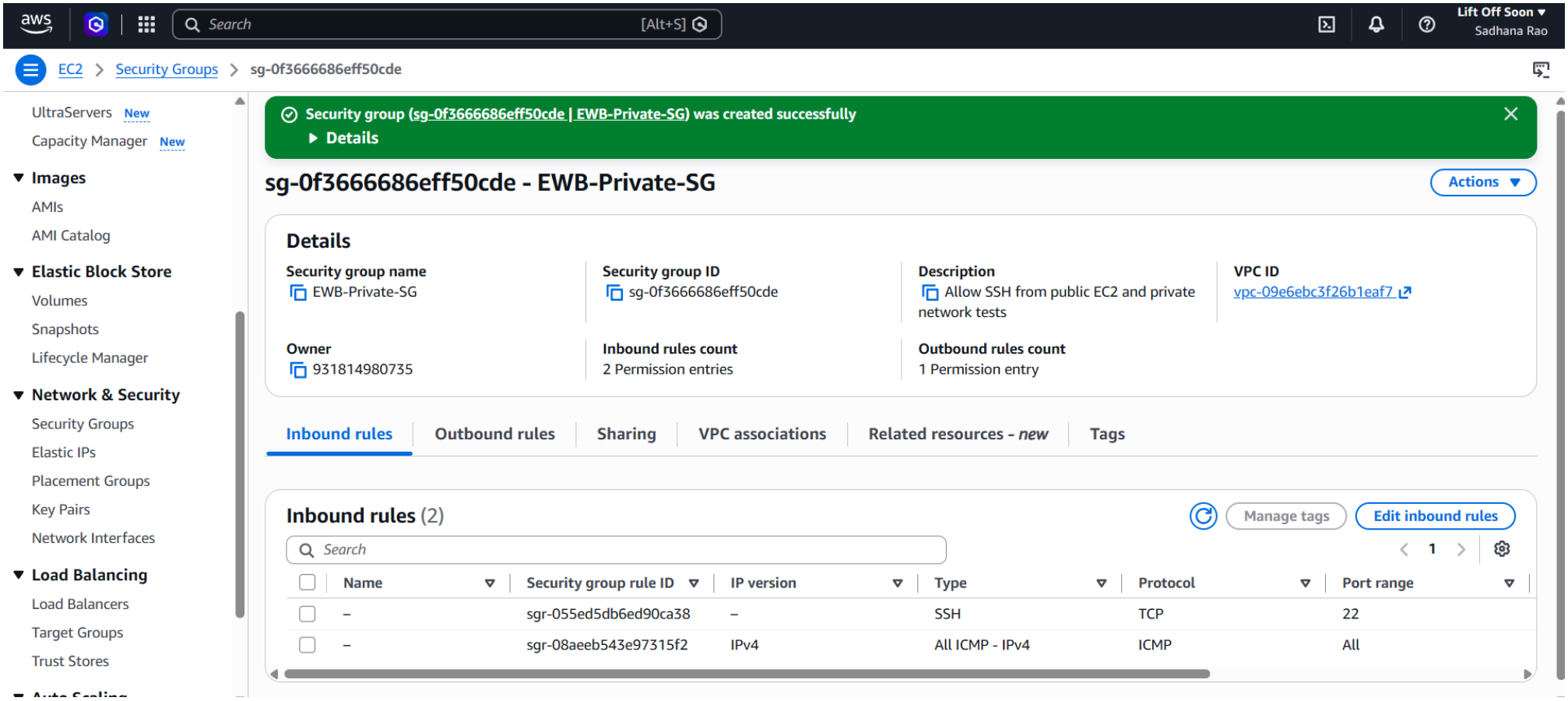
06. Configure the Public EC2 Security Group

Set inbound access for the public EC2 instance, including SSH from the administrator's IP.



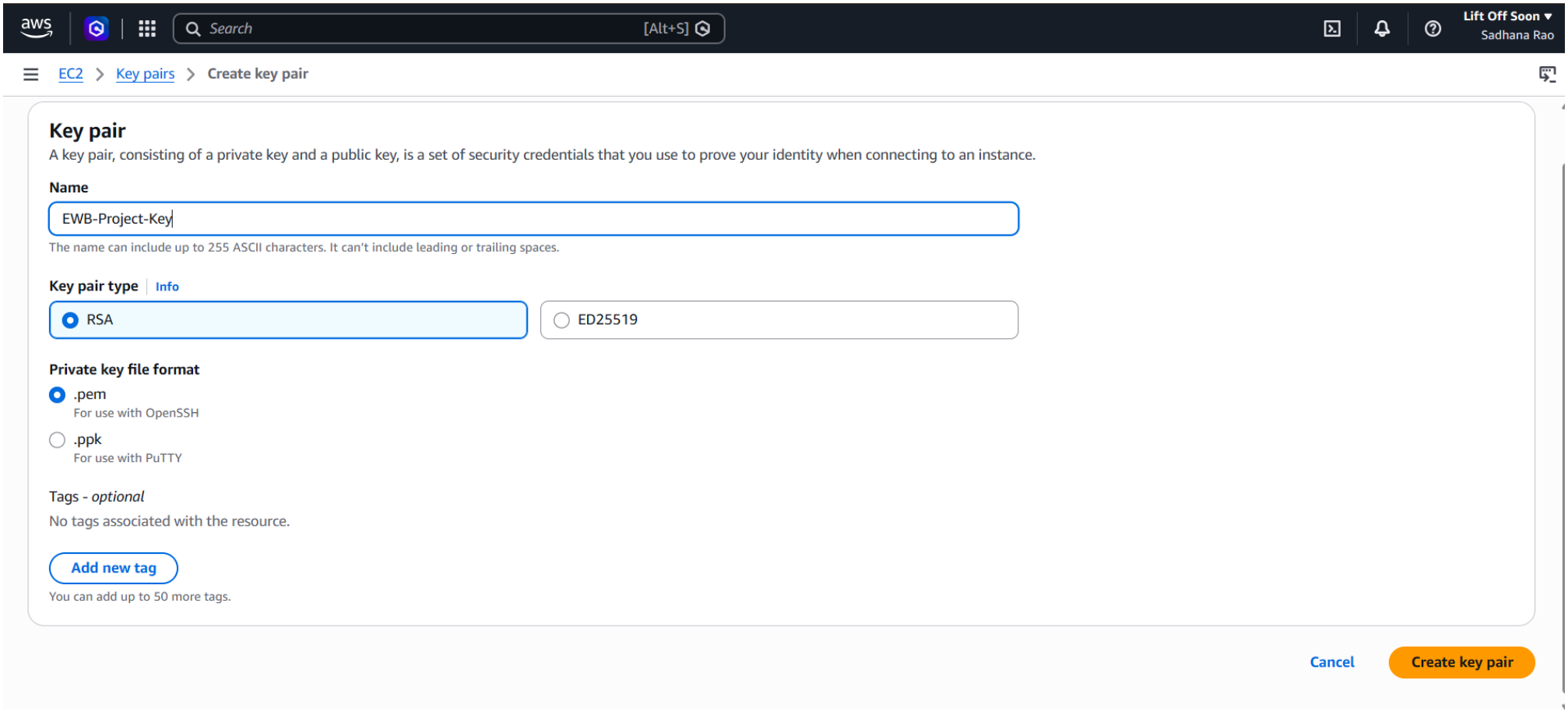
07. Configure the Private EC2 Security Group

Allow SSH from the public jump-host security group and ICMP for the project tests.



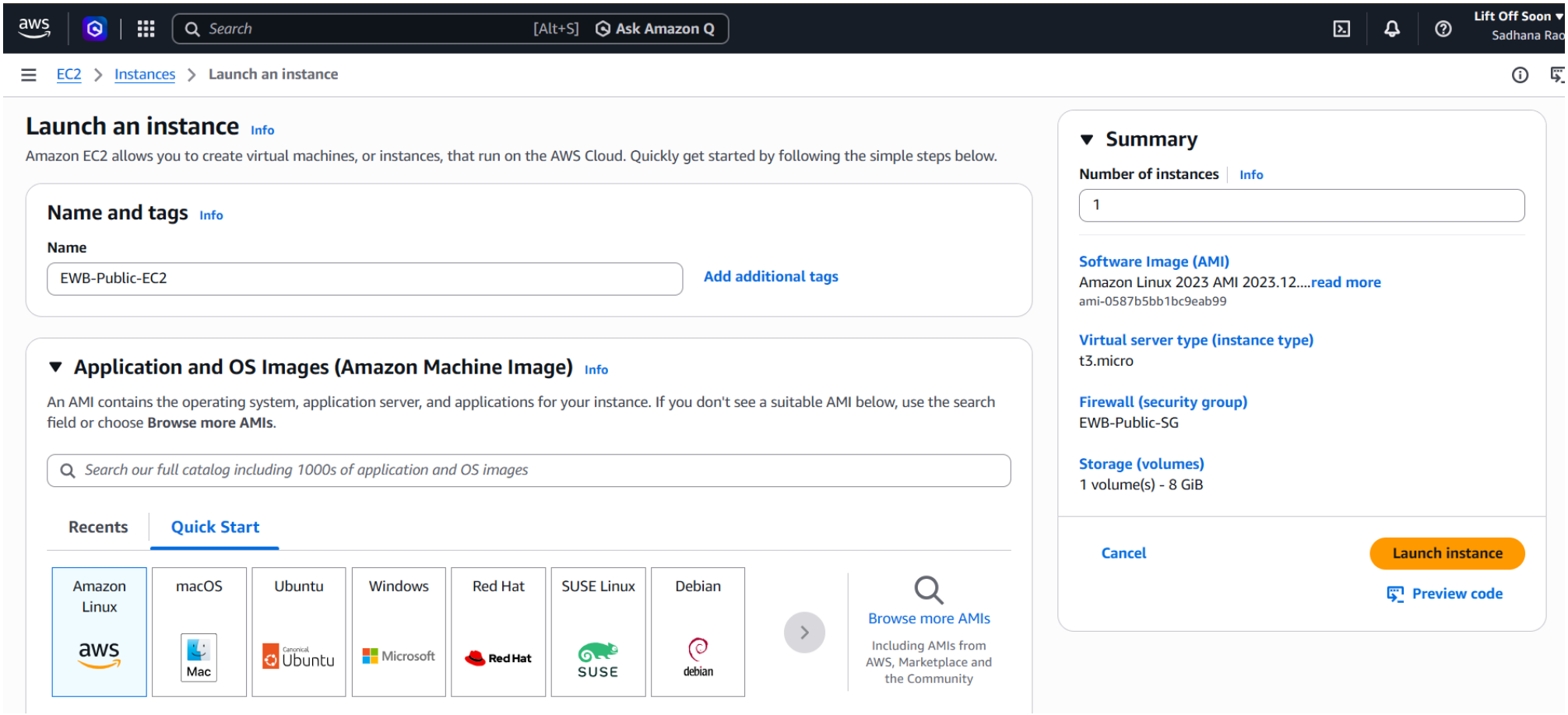
08. Create the EC2 Key Pair

Create the key pair used to securely connect to the EC2 instances.



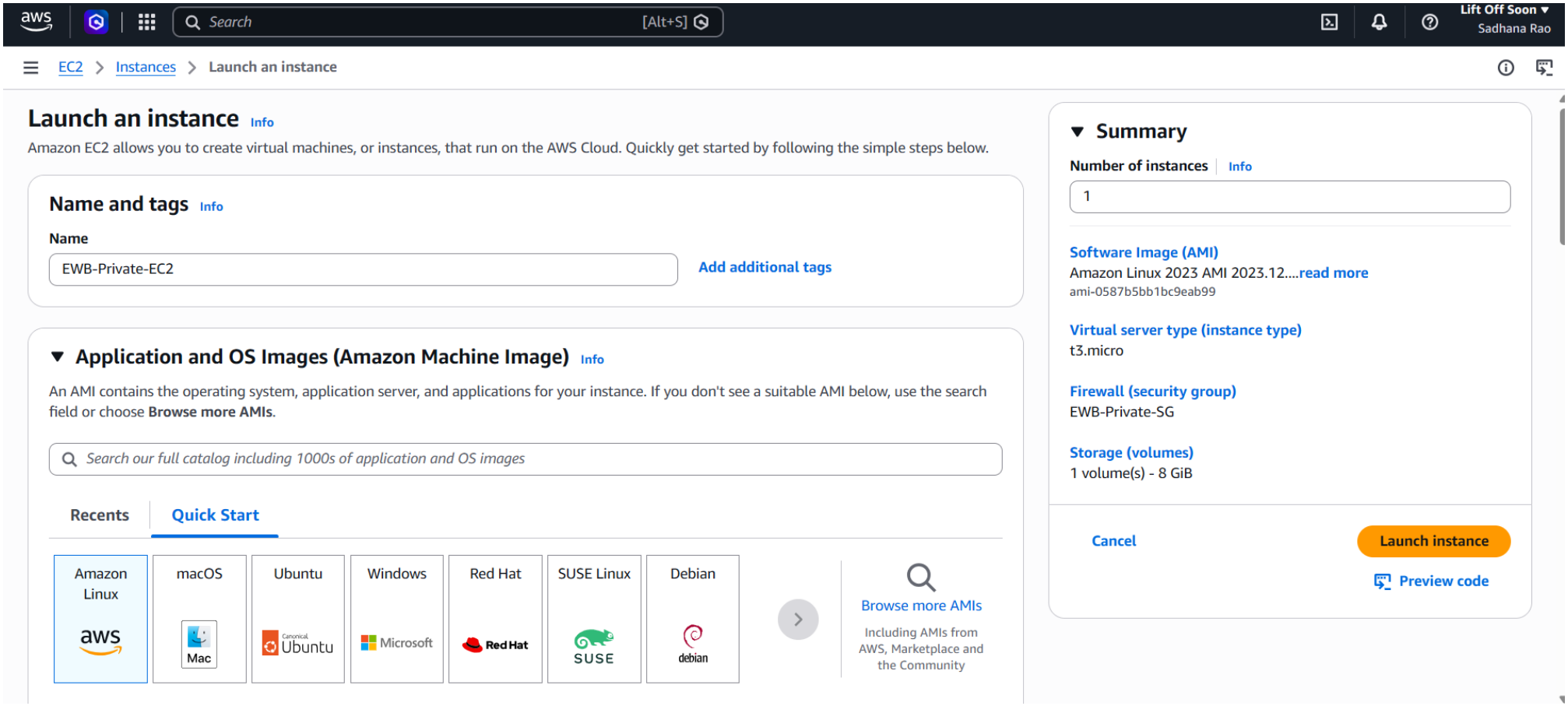
09. Launch the Public EC2 Instance

Launch EWB-Public-EC2 in the public subnet with a public IPv4 address.



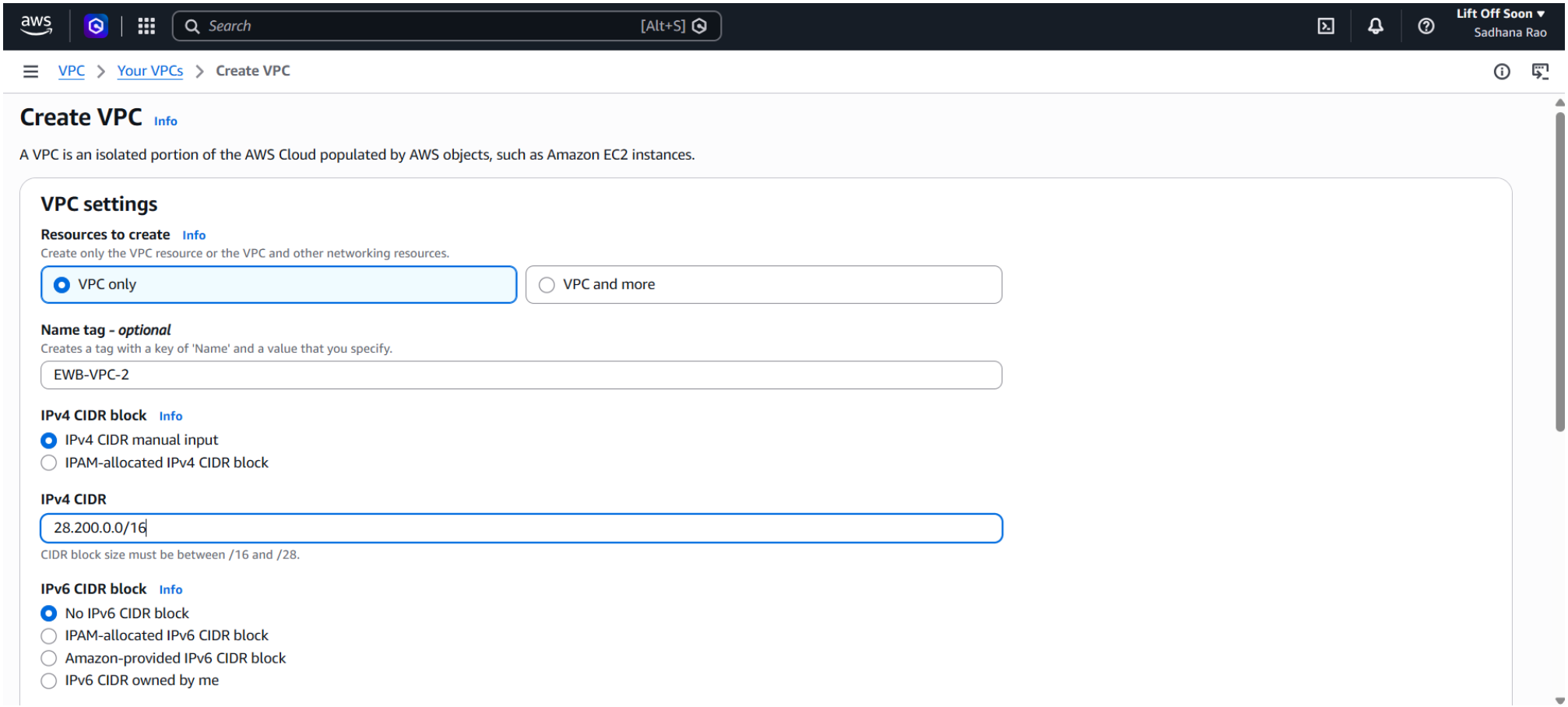
10. Launch the Private EC2 Instance in VPC 1

Launch EWB-Private-EC2 in the private subnet without a public IP address.



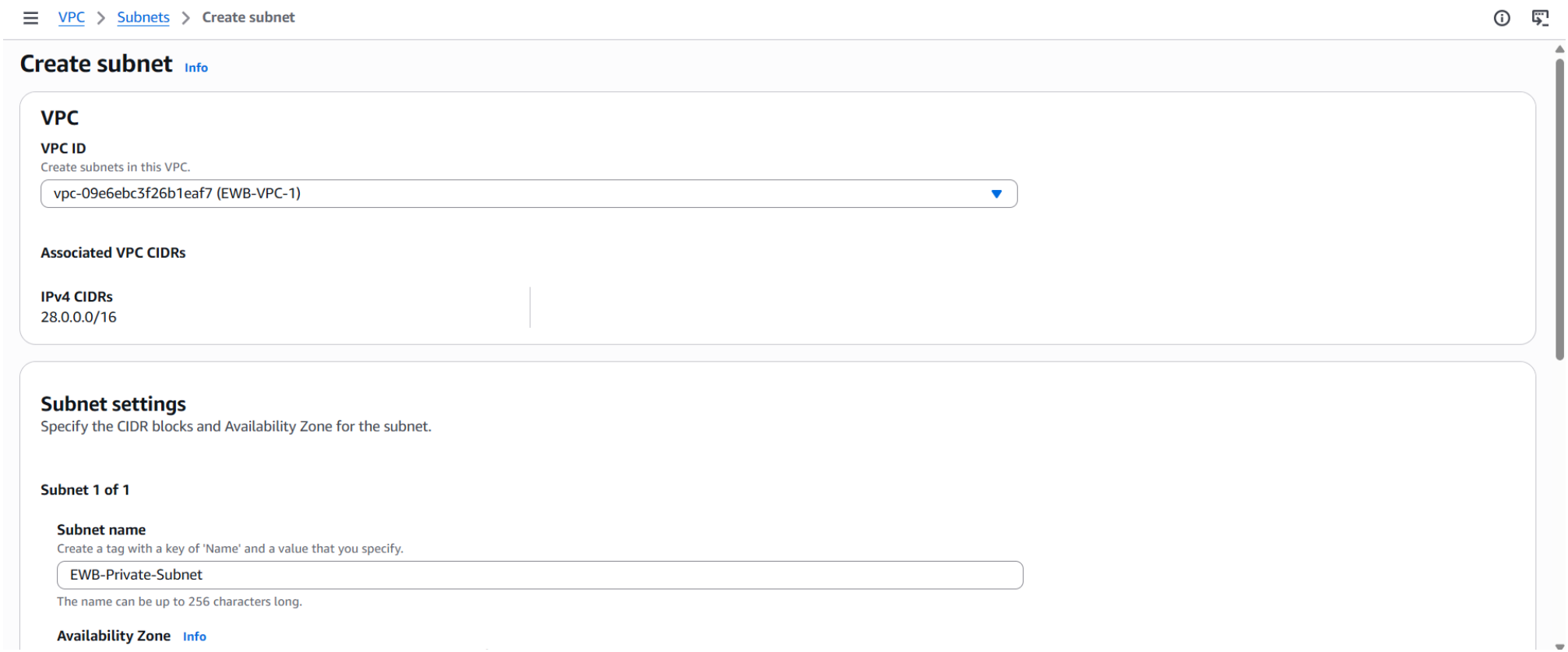
11. Create VPC 2 — EWB-VPC-2

Create the second VPC with CIDR block 28.200.0.0/16.



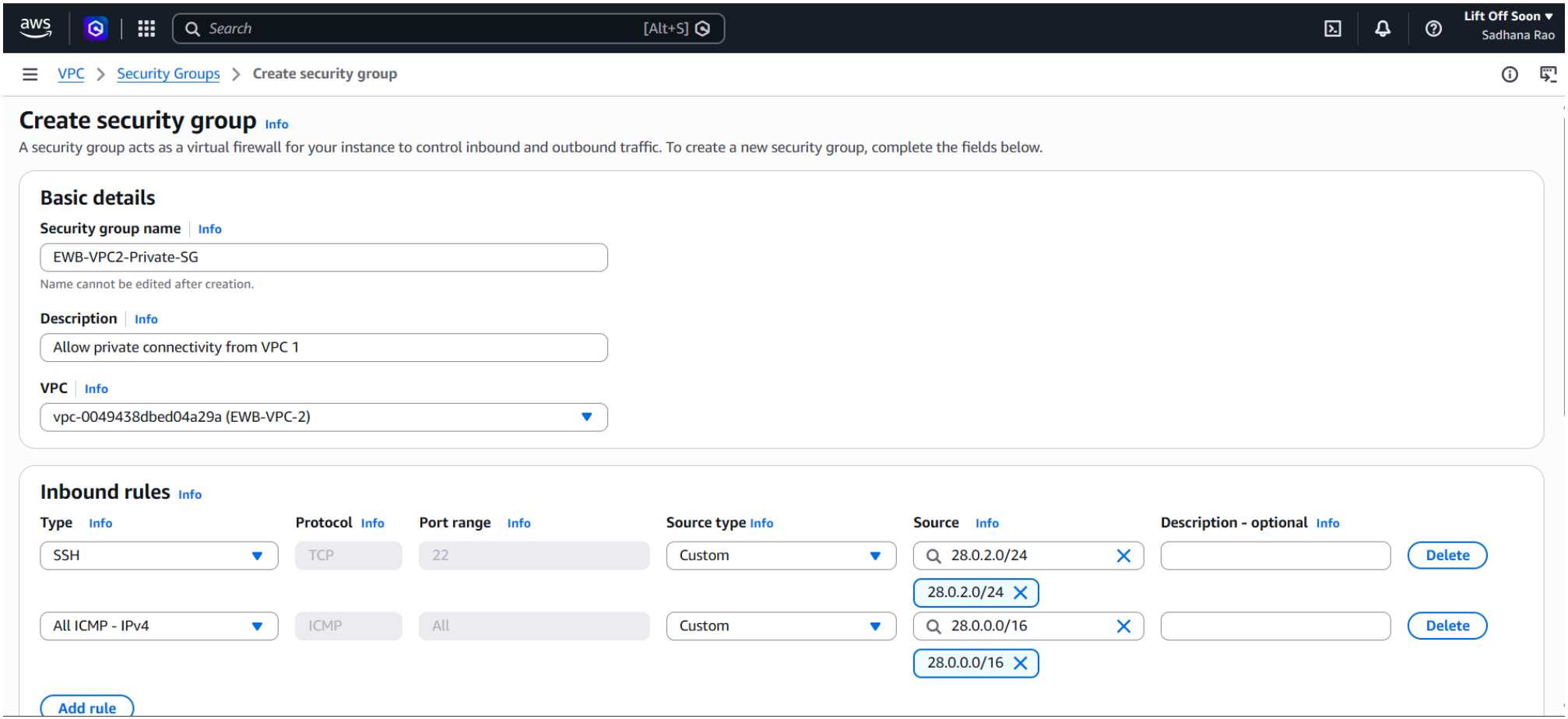
12. Create the Private Subnet in VPC 2

Configure EWB-VPC2-Private-Subnet with CIDR block 28.200.1.0/24.



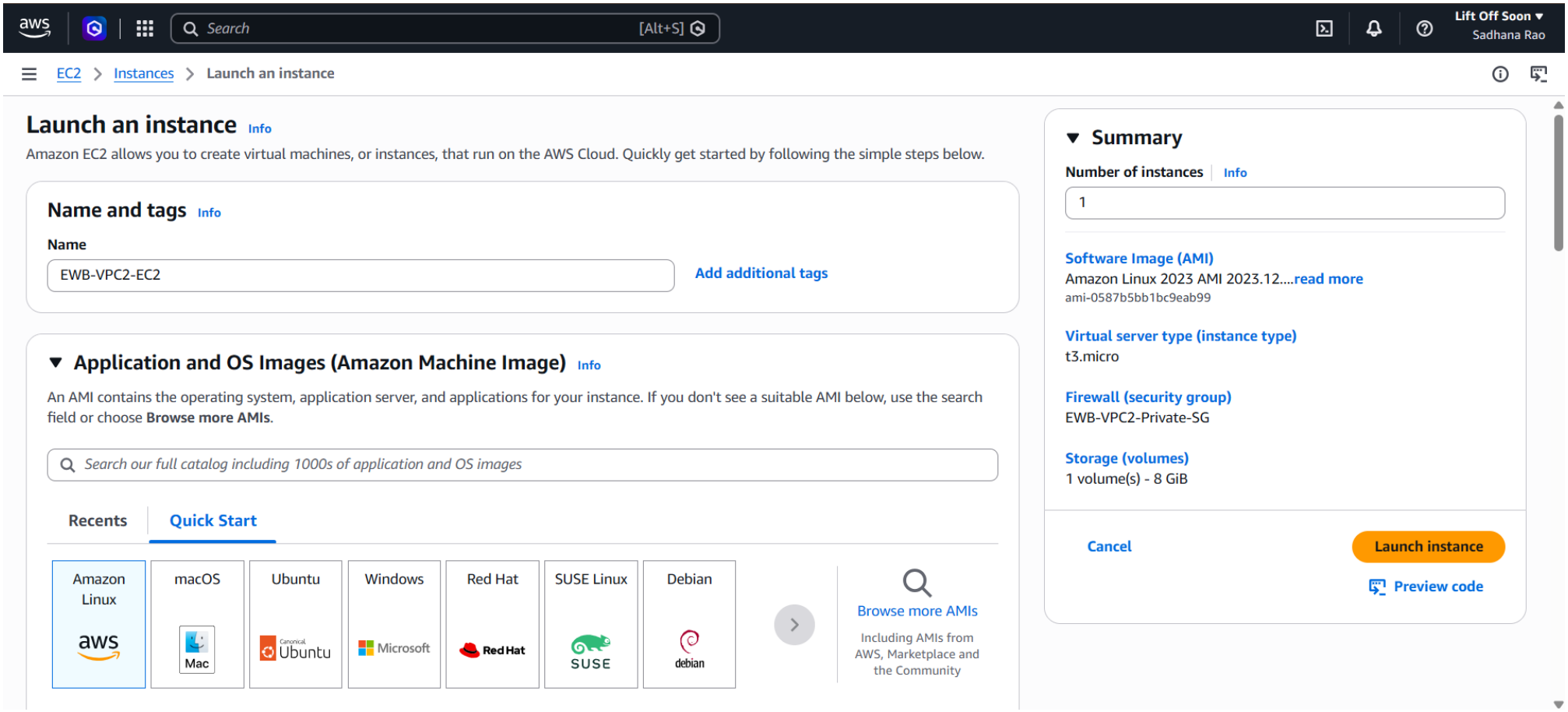
13. Configure the VPC 2 Private Security Group

Allow the required SSH and ICMP traffic for private connectivity tests.



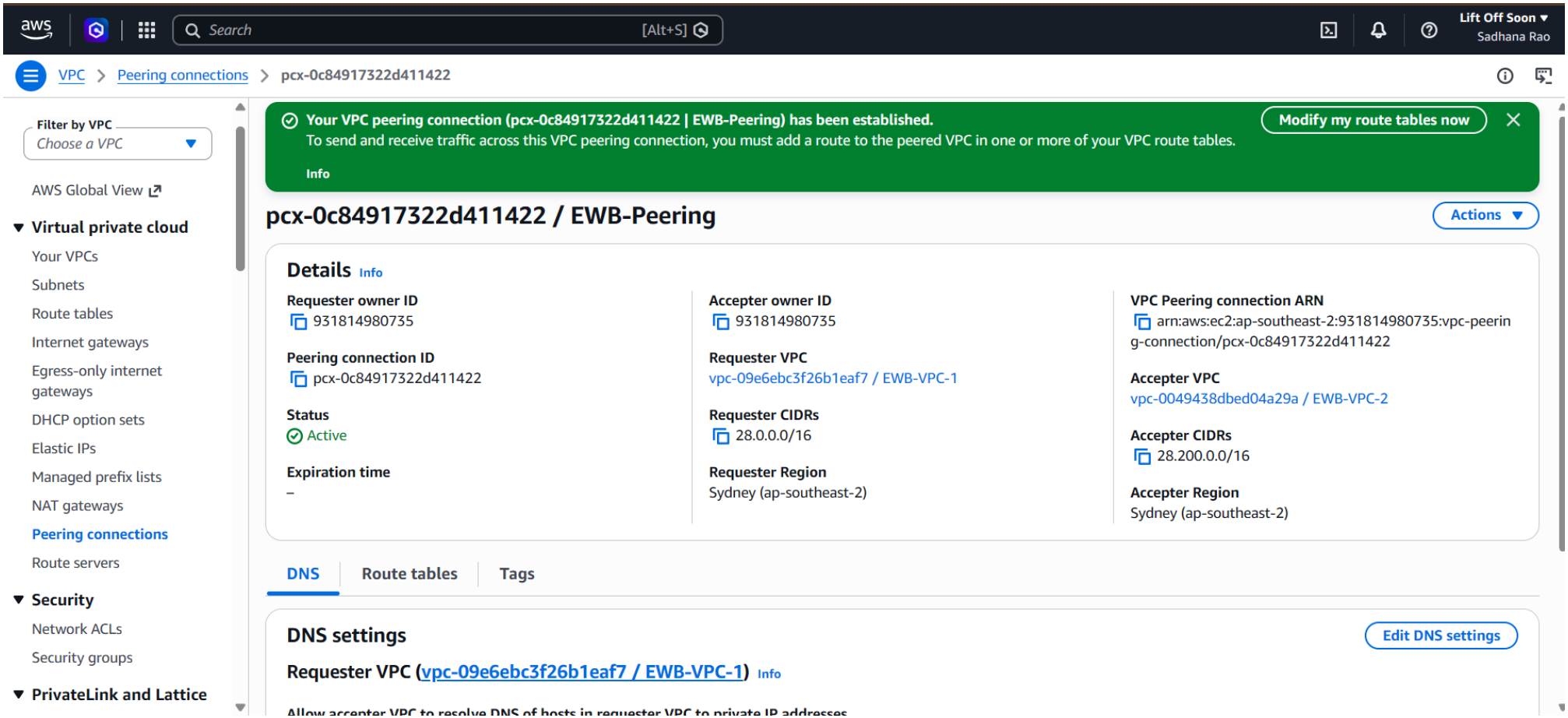
14. Launch the Private EC2 Instance in VPC 2

Launch EWB-VPC2-EC2 in the VPC 2 private subnet without a public IP address.



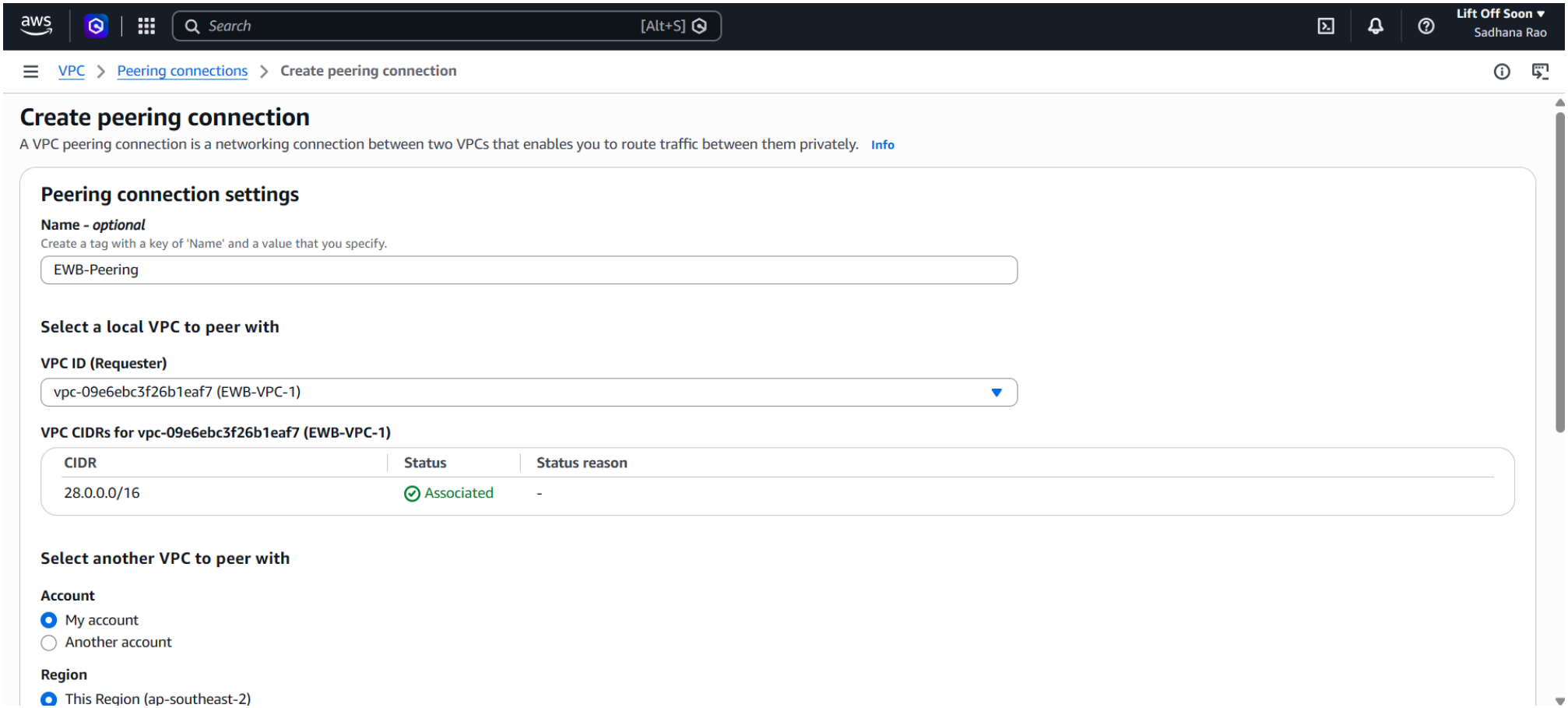
15. Create the VPC Peering Connection

Create a peering connection between EWB-VPC-1 and EWB-VPC-2.



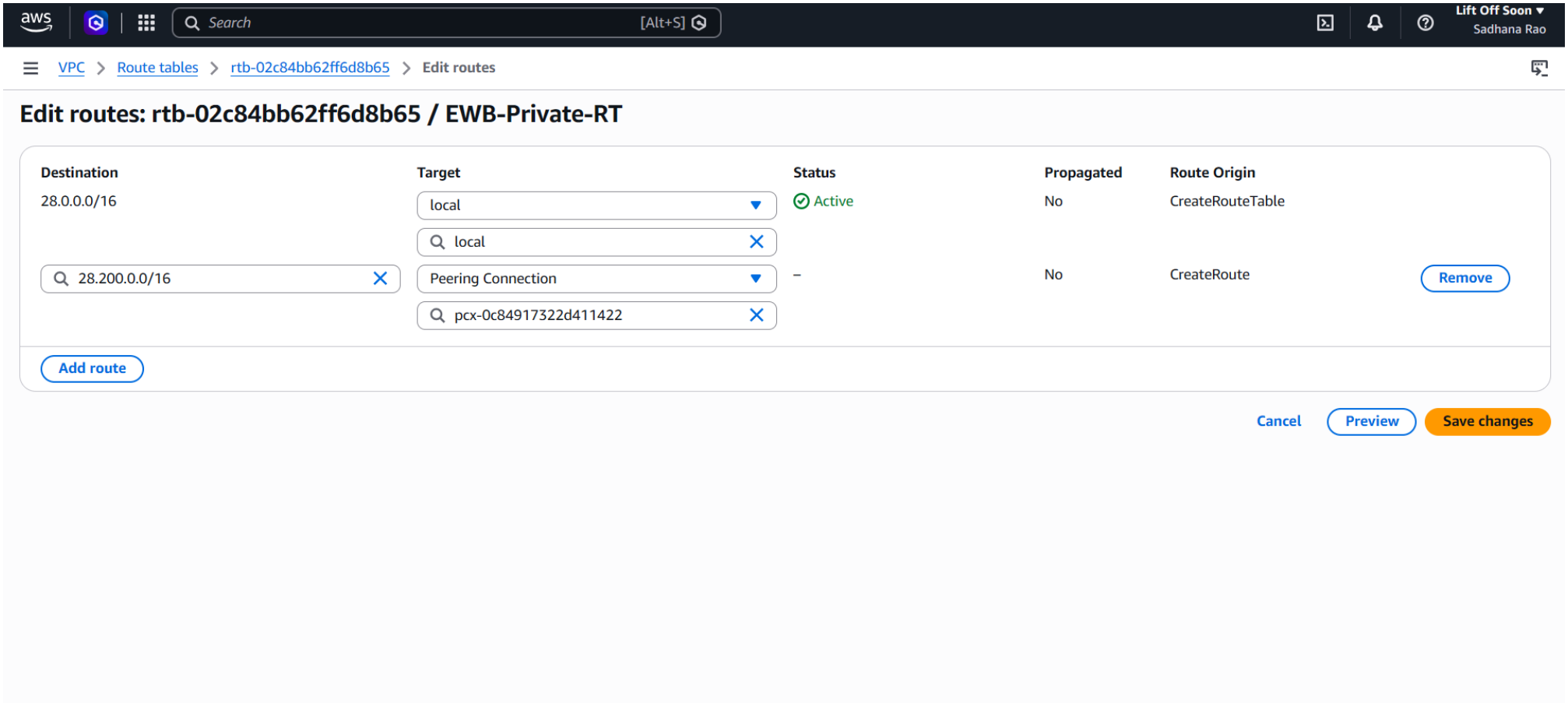
16. Verify the VPC Peering Connection

Check that the peering connection is available/active.



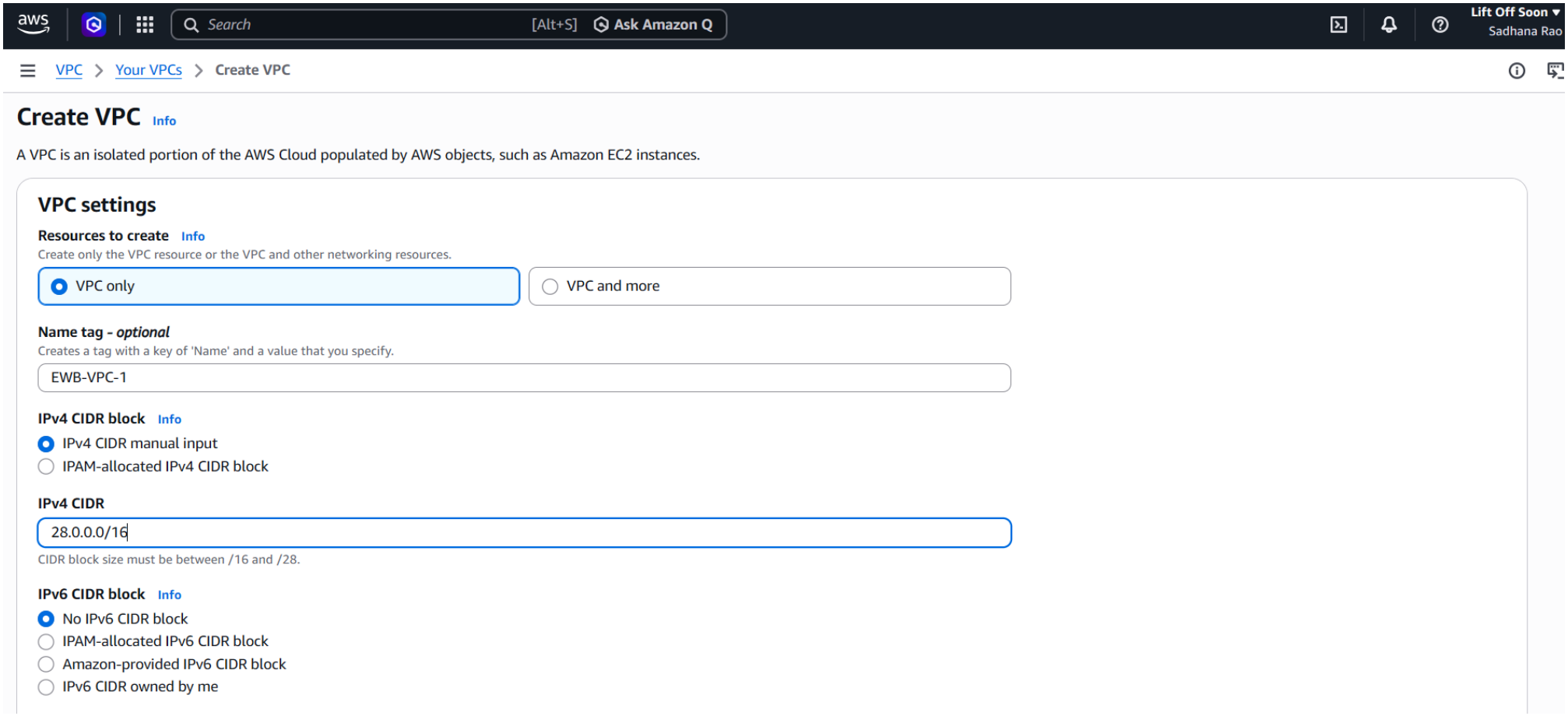
17. Add the Peering Route to the Route Table

Route traffic for 28.200.0.0/16 through the VPC peering connection.



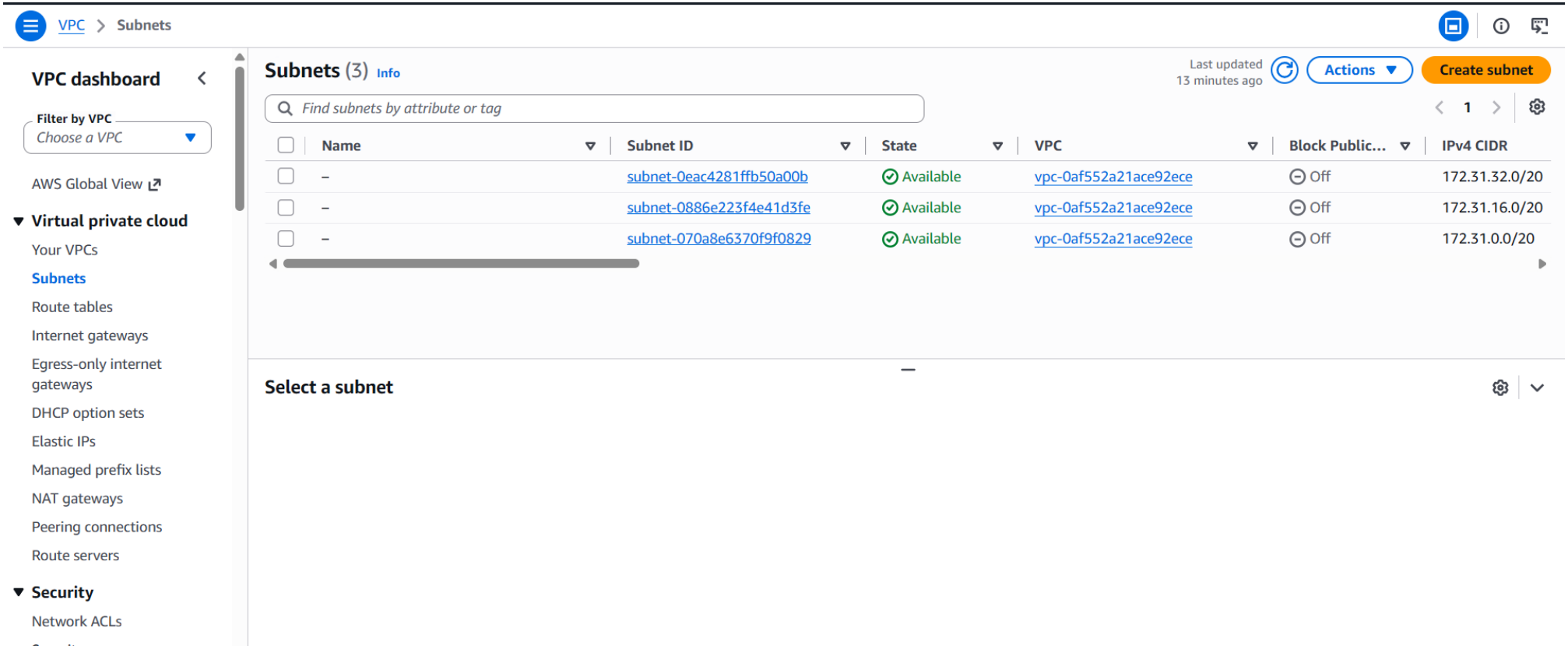
18. Review VPC 1 Configuration

Review the VPC 1 resource settings before completing network connectivity.



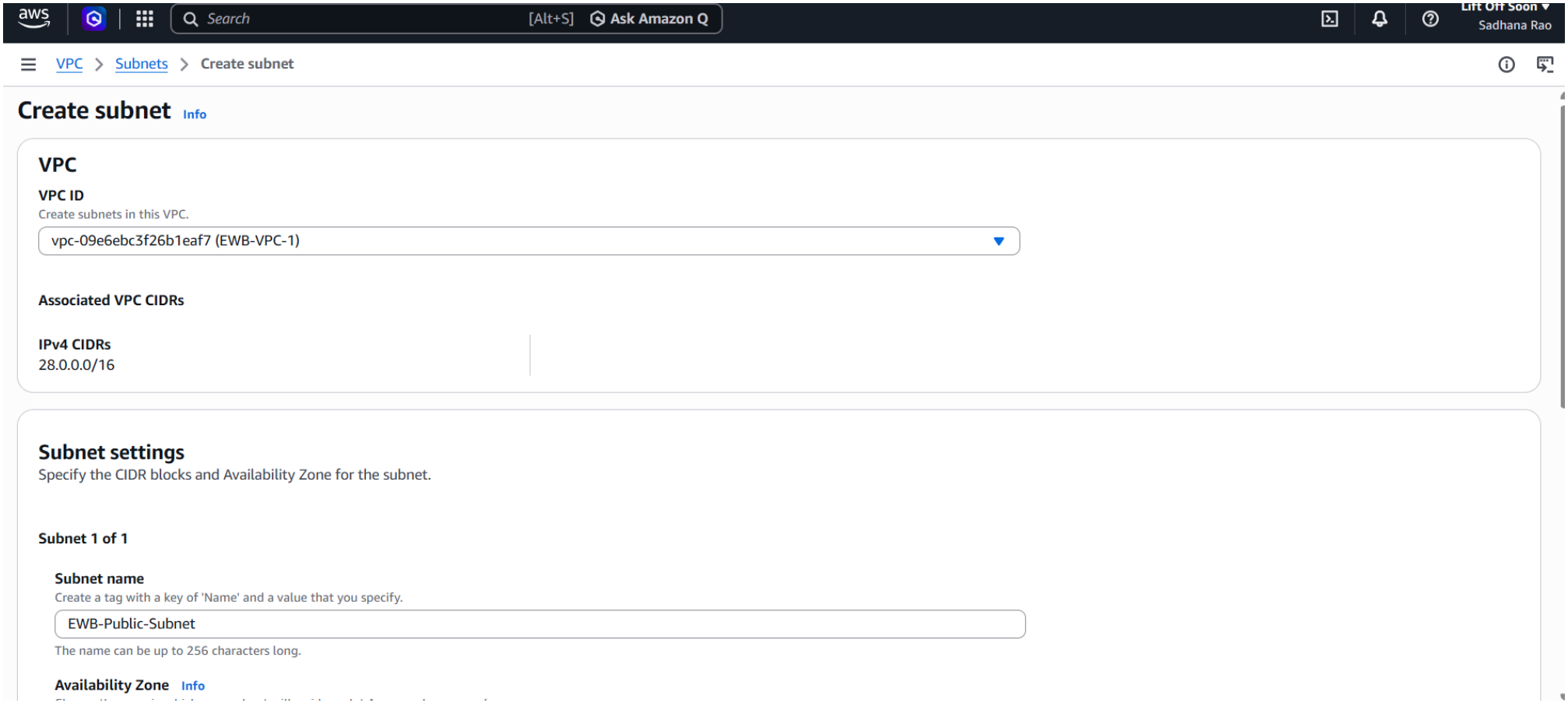
19. Review the Subnet Configuration

Verify the created subnet and its CIDR allocation.



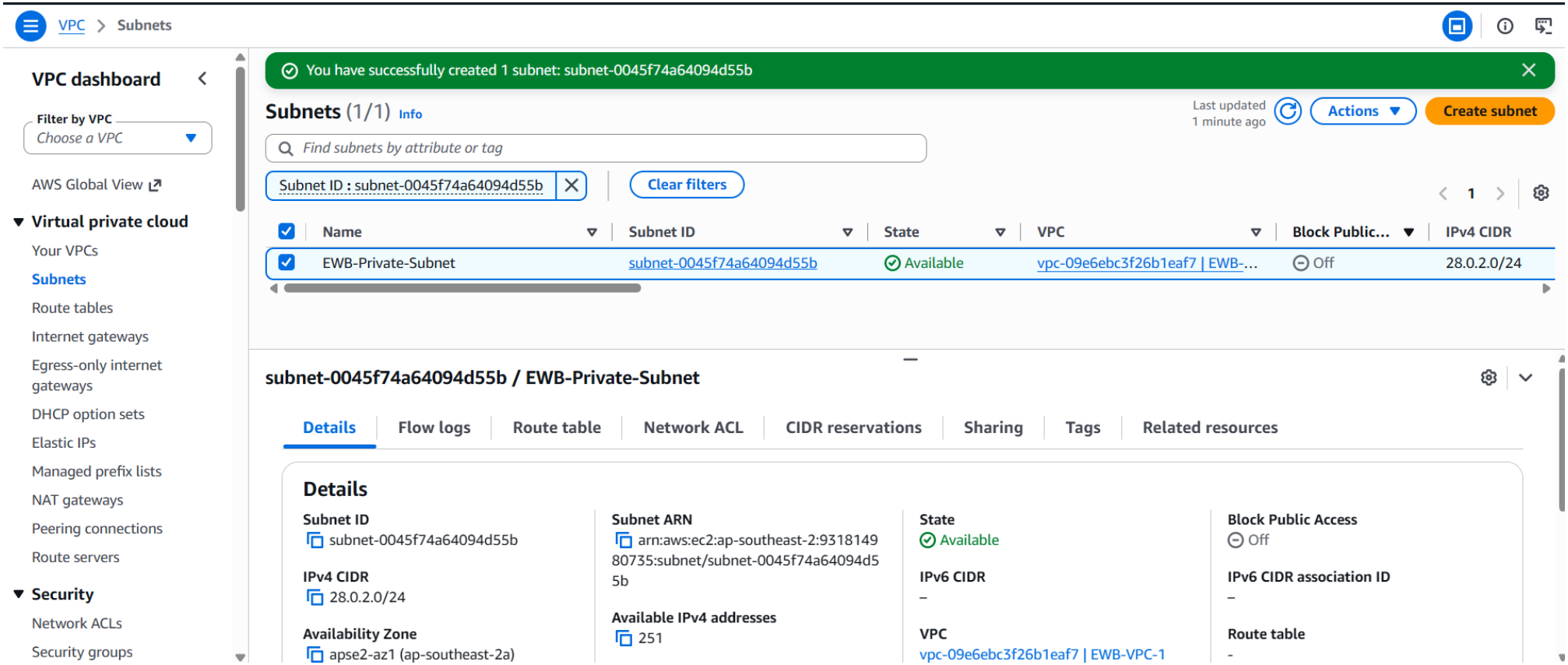
20. Verify the Private Subnet

Confirm the private subnet resource is present in the VPC.



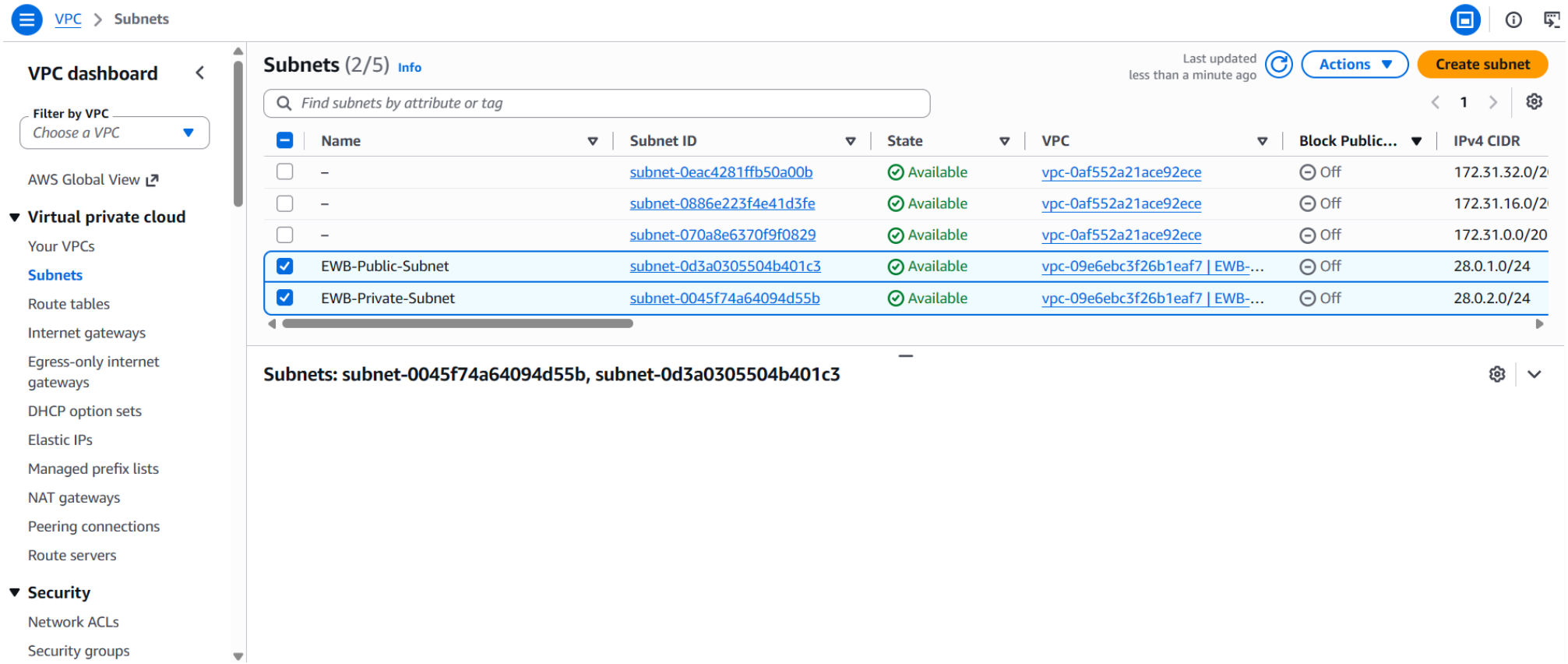
21. Configure the VPC 2 Route

Add the return route from VPC 2 to the VPC 1 CIDR through peering.



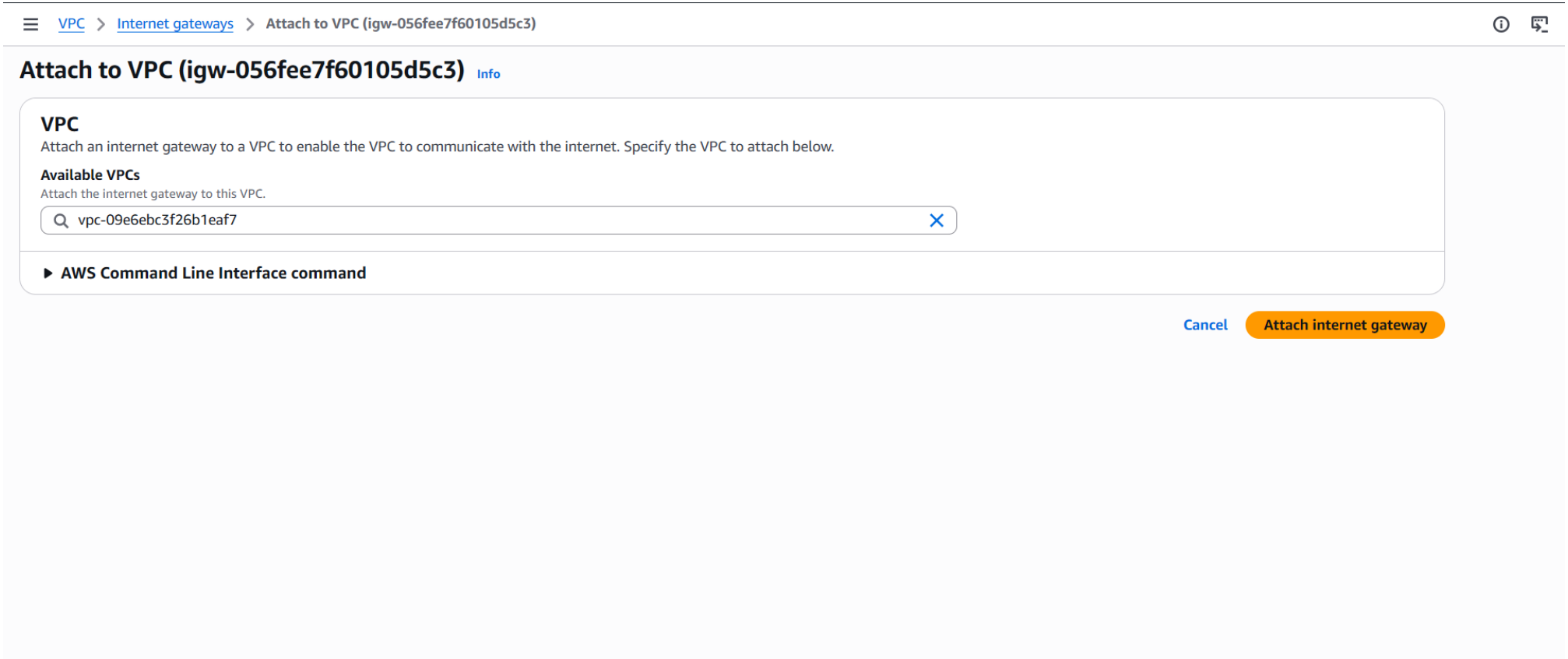
22. Review EC2 Network Settings

Verify instance placement and network configuration.



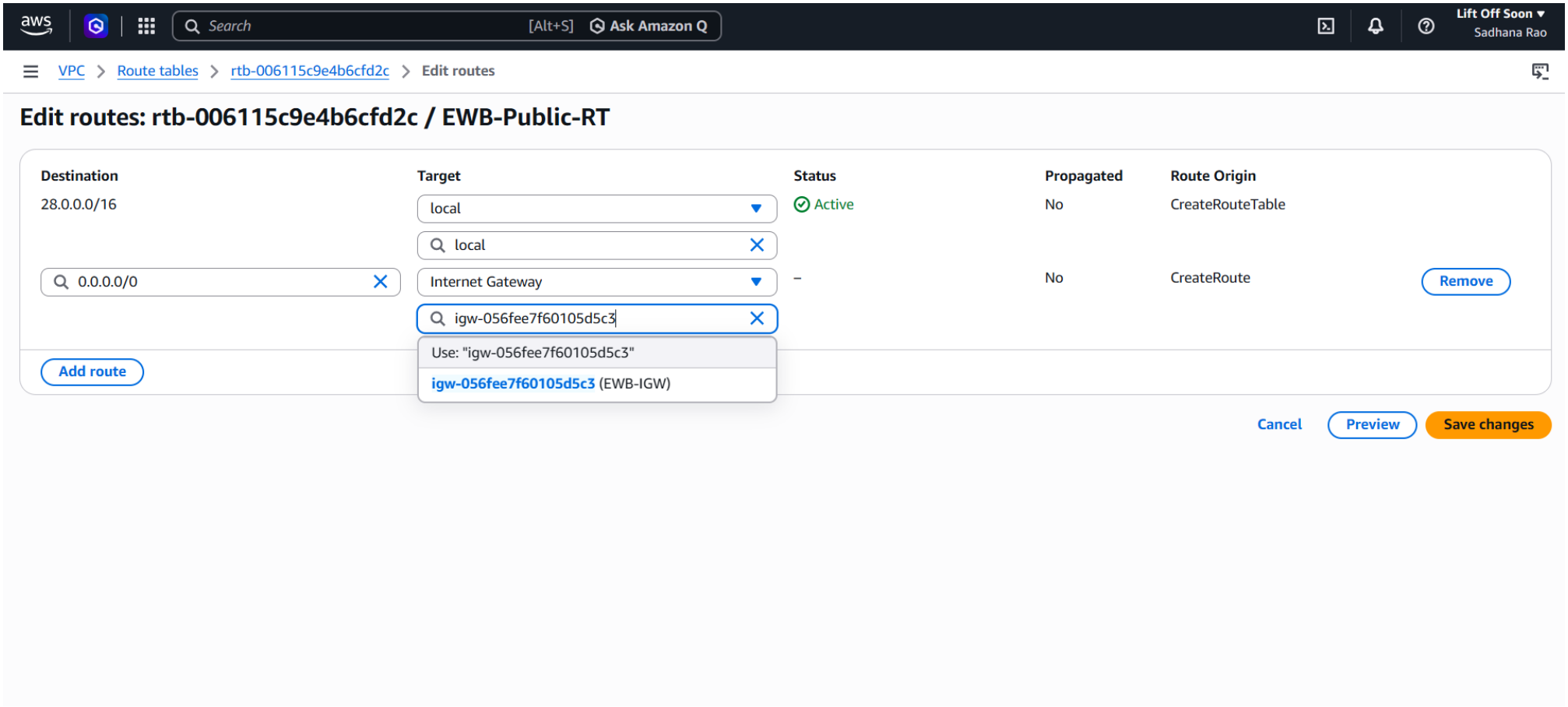
23. Review the Peering Route Settings

Confirm the relevant route destination and peering target.



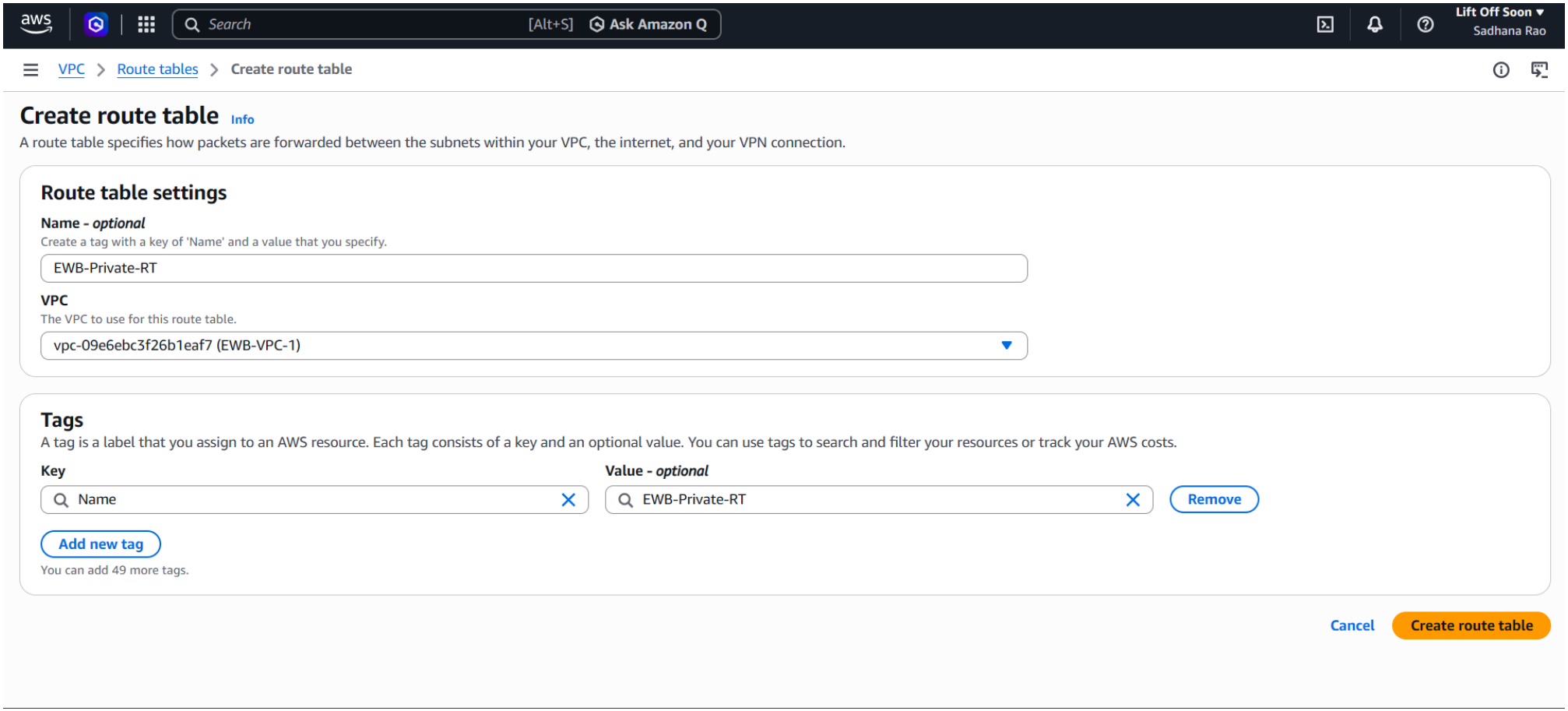
24. Verify Security Group Rules

Review the configured inbound rules used for private connectivity.



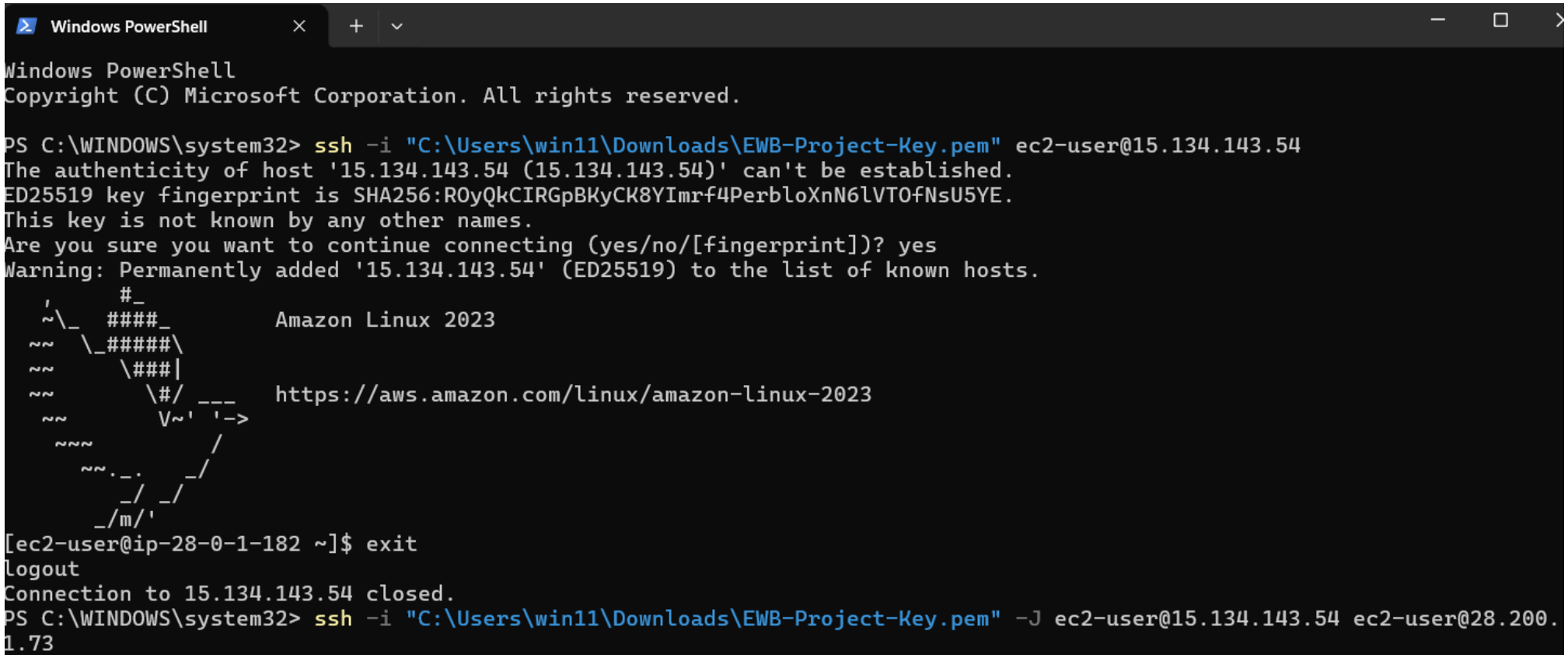
25. Verify the VPC Peering Route

Check the route table entry used to reach the peer VPC.



26. Connect to the Private EC2 Instance Using SSH

Use the public EC2 instance as a jump host to reach the private instance.



28. Test Connectivity Across VPC Peering

Ping the private IP address of the VPC 2 instance from the private instance in VPC 1.

```
Windows PowerShell
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PS C:\WINDOWS\system32> ssh -o IdentitiesOnly=yes -i "C:\Users\win11\Downloads\EWB-Project-Key.pem" -o "ProxyCommand=ssh -o IdentitiesOnly=yes -i C:\Users\w
in11\Downloads\EWB-Project-Key.pem -W %h:%p ec2-user@15.134.143.54" ec2-user@28.0.2.180
The authenticity of host '28.0.2.180 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:7Ea/HtvEMs9nPBBnxf5V0JsbuwhYrY4cP2pv2cAfHeM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '28.0.2.180' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-28-0-2-180 ~]$ ping -c 4 28.200.1.73
PING 28.200.1.73 (28.200.1.73) 56(84) bytes of data.
64 bytes from 28.200.1.73: icmp_seq=1 ttl=127 time=0.477 ms
64 bytes from 28.200.1.73: icmp_seq=2 ttl=127 time=0.281 ms
64 bytes from 28.200.1.73: icmp_seq=3 ttl=127 time=0.293 ms
64 bytes from 28.200.1.73: icmp_seq=4 ttl=127 time=0.268 ms

--- 28.200.1.73 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3117ms
rtt min/avg/max/mdev = 0.268/0.329/0.477/0.085 ms
[ec2-user@ip-28-0-2-180 ~]$
```

29. Confirm Successful Ping Results

The terminal output shows 4 packets transmitted, 4 received, and 0% packet loss.

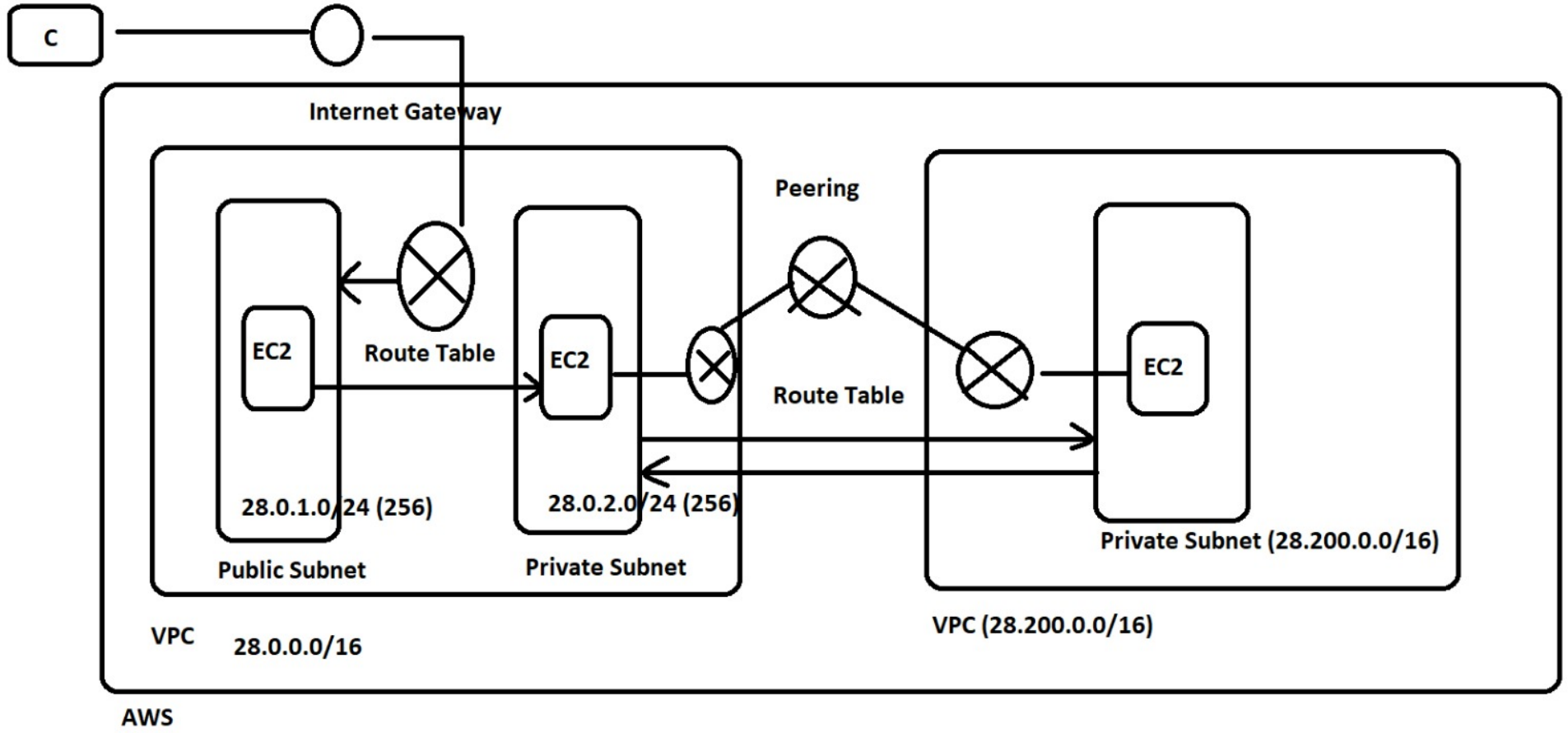
```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> ssh -o IdentitiesOnly=yes -i "C:\Users\win11\Downloads\EWB-Project-Key.pem" -o "ProxyCommand=ssh -o IdentitiesOnly=yes -i C:\Users\win11\Downloads\EWB-Project-Key.pem -W %h:%p ec2-user@15.134.143.54" ec2-user@28.0.2.180
The authenticity of host '28.0.2.180 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:7Ea/HtvEMs9nPBBnxf5V0JsbuwhYrY4cP2pv2cAfHeM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '28.0.2.180' (ED25519) to the list of known hosts.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-28-0-2-180 ~]$ ping -c 4 28.200.1.73
PING 28.200.1.73 (28.200.1.73) 56(84) bytes of data.
64 bytes from 28.200.1.73: icmp_seq=1 ttl=127 time=0.477 ms
64 bytes from 28.200.1.73: icmp_seq=2 ttl=127 time=0.281 ms
64 bytes from 28.200.1.73: icmp_seq=3 ttl=127 time=0.293 ms
64 bytes from 28.200.1.73: icmp_seq=4 ttl=127 time=0.268 ms

--- 28.200.1.73 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3117ms
rtt min/avg/max/mdev = 0.268/0.329/0.477/0.085 ms
[ec2-user@ip-28-0-2-180 ~]$
```

30. Project Network Overview

Overview of the intended two-VPC design with public and private resources and a peering link.



Conclusion

This project provided practical experience in AWS VPC networking, subnetting, route tables, security groups, EC2 access, and VPC peering. The connectivity test showed successful communication between the private instances across the peered VPCs.